MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10ALP19

This specification sheet is approved for use by U.S. Army Electronics Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, special deflection structure for minimum deflection defocusing, aluminized screen.

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	ed	Rg	Zd
Unit:	V	Vdc	Vdc	Vdc	v	Meg	Meg (see note 1)
Maximum:	6.9	0	4,000	10,000	1,200	1.5	5.0
Minimum:	5.7	-300	---	5,000	---	---	---
TEST CONDITIONS: (see note 2)	6.3	Adjust Focus		8,000	---	---	---

GENERAL:

Preproduction sample approved inspection - Required (see note 3)

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Preproduction sample approval inspection	6					
1141	Pressure (implosion)	-		---	---	---	---
1331	Direct - interelectrode capacitance	-	Grid 1 to all	Cg1	---	12.5	pF
			D1 to D2	C1D2	---	7.0	pF
			D3 to D4	C3D4	---	3.0	pF
			D1 to all	CD1	---	14.5	pF
			D2 to all	CD2	---	14.5	pF
			D3 to all	CD3	---	8.0	pF
			D4 to all	CD4	---	8.0	pF
5216	Cathode illumination	-		---	---	---	---
5248	Deflection - factor uniformity	-		---	---	1.5	%
5111	Vibration	-		---	---	1	mm
5115	Shock	11		---	---	---	---
1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
5101	Angle between traces	-		---	89	91	Degrees
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross" (electrostatic deflection)	7	Beam current = 2.5 µAdc	---	---	---	---
5216	Stray light emission (conventional types)	-	Eb = 10,000 Vdc	---	---	---	---
5223	Modulation	7	Beam current = 2.5 µAdc	ΔEc1	---	20	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	20	mm
5241	Grid-cutoff voltage	-		Ec1	-155	-205	Vdc
5246	Focusing voltage at cutoff	-		Eb1	2,250	3,100	Vdc
5231	Spot displacement (leakage)	-		---	---	10	mm
5251	Grid No. 1 leakage current	-		---	---	---	---
---	Spot examination	8		---	---	---	---
---	Minimum useful scan	7	Focused trace; beam current = 2.5 µAdc	---			
	1D2	-		---	9	---	Inches
	3D4	-		---	9	---	Inches
---	Screen charging	9		---	---	---	---
---	Deflection defocusing	10		---	---	1:3	Ratio
5103	Pattern distortion	-		---	---	2	%

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 - Continued						
1101	Secureness of base, cap, or insert	-		---	---	---	---
1111	Base pin solder depth	-		---	---	---	---
1105	Permanence of marking	4, 5		---	---	---	---
	Quality conformance inspection, part 2						
5101	Neck and base alinement (electrostatic types)	-		---	---	1.5	Degrees
1301	Heater current	-		If	540	660	mA
5101	Base alinement (electrostatic types)	-	+1D2, index pin	---	---	---	---
5201	Electrode current (anode No. 1)	-		Ib1	-15	+10	µAdc
5221	Screens	7		---	---	---	---
5226	Line width A (electrostatic deflection)	7	Beam current = 2.5 µAdc	---	---	0.30	mm
5226	Line width B (electrostatic deflection)	7	Beam current = 2.5 µAdc	---	---	0.42	mm
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
5248	Deflection factor	-	1D2	---	155	189	Vdc/inch
5248	Deflection factor	-	3D4	---	151	185	Vdc/inch
	Quality conformance inspection, part 3						
---	Life test	7, 11	Group C; beam current = 2.5 µAdc; Eb2 = 10,000 Vdc; t = 500 hours(min)	---	---	---	---
---	Life-test end points:	-					
5223	Modulation	-		ΔEc1	---	20	Vdc
5226	Line width A	-		---	---	0.30	mm
5226	Line width B	-		---	---	0.42	mm
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
5216	Stray light emission (conventional types)	-		---	---	---	---

NOTES:

1. It is recommended that the deflection electrode circuit resistances be approximately equal.
2. The P19 screen may be permanently damaged if the current density is permitted to rise too high. To prevent burning, minimum beam current densities should be employed.

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NOTES: - Continued

3. The preproduction sample approval requirements in U.S. Army Electronics Command Drawing SC-A-466008 hereby replace any qualification requirements referable to the product covered herein. However, the requirements of 3.1.2.1, 3.1.2.2, 3.1.3, and 4.4 in Drawing SC-A-466008 shall be considered not applicable to the product covered herein.

   (Copies of U.S. Army Electronics Command Drawing SC-A-466008, 19 March 1965, "Preproduction Sample Approval In Lieu of Qualification Requirements In Specifications For Semiconductor Devices And Electron Tubes" may be obtained from the procuring activity or as directed by the contracting officer.)

4. Marking shall be in accordance with MIL-E- 1. If any specification requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only. (See figure 1 herein.)

5. The following additional marking requirement shall apply: A line, of adequately contrasting color to the base-material color, shall be drawn on the tube base parallel to the tube axis; this line shall be alined with and on the same side as the index pin.

6. Although all tests covered herein are required to be performed during preproduction sample approval inspection, the seven tests listed under this caption are performed, normally, during preproduction sample approval only.

7. Beam current shall be measured by applying a sufficient voltage to either D1 or D2 to deflect a 1 by 1-inch (25.4 by 25.4 mm) raster until the raster cannot be seen on the screen. Grid No. 1 bias voltage shall be adjusted so that either D1 or D2 will collect 2.5 microamperes.

8. With the focused spot just above visual cutoff, examine the spot for any abnormalities. The spot shall be of uniform brightness and shall not show any indication of a dual spot. This may be noted as an astigmatic spot of nonuniform brightness, or a spot the extremities of which give the appearance of bright cores, with the center of the spot appearing dark.

9. Visually check the internal aluminum coating in the area between the edge of the faceplate and internal aquadag; it shall appear uniform and nearly opaque. If not, perform the following test: With a 1.5 by 1.5-inch (38.1 by 38.1 mm) slightly defocused raster, per method 5221, in the center of the screen, and with Eb2 = 10,000 volts, and with a beam current of 15 microamperes, there shall be no evidence of any doubling of the raster lines (venetian-blind effect), or a sawtooth effect at the edges of the raster. (Any doubling of the raster lines should be intermittent only, giving the appearance of being pulsed.) Repeat on the edges of the screen in the following positions:

   a. 45° clockwise from the plane of pin No. 1.

   b. 135° clockwise from the plane of pin No. 1.

   c. 225° clockwise from the plane of pin No. 1.

   d. 315° clockwise from the plane of pin No. 1.

   These tests shall be made quickly to prevent burning of the screen; apply the grid drive momentarily.

10. The ratio of the line width of a 2-inch (50.8 mm) long trace, adjusted for best center focus, to the line width measured on the faceplate axis when the trace is deflected perpendicularly plus or minus 4 inches (101.6 mm) from the face center by a balanced dc voltage, shall not be greater than the value specified herein.

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NOTES: - Continued

11. Tubes subjected to the following destructive tests are not to be delivered on the contract: Shock; life test.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

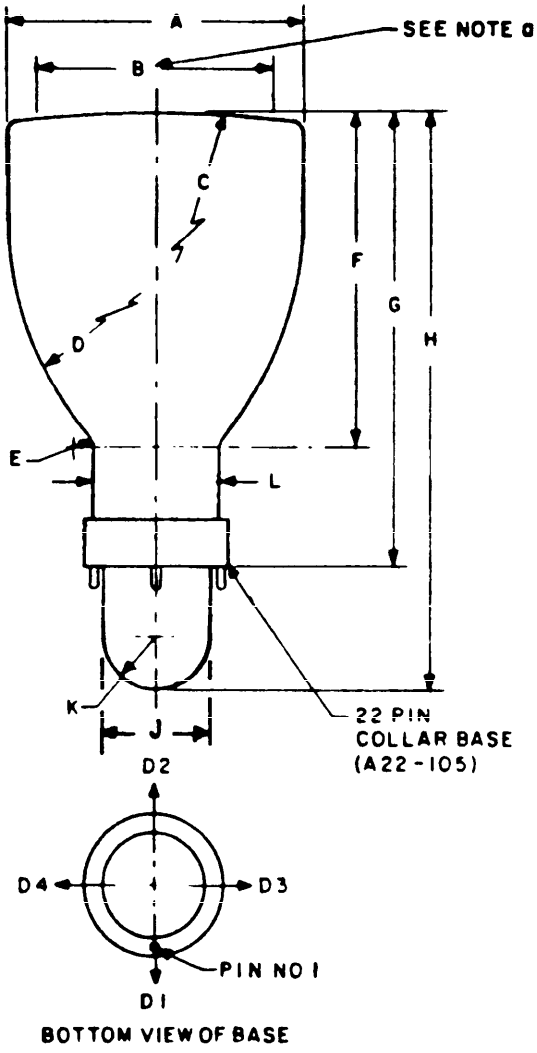
User activities:
Army - ME, AV

Preparing activity:
Army - ER

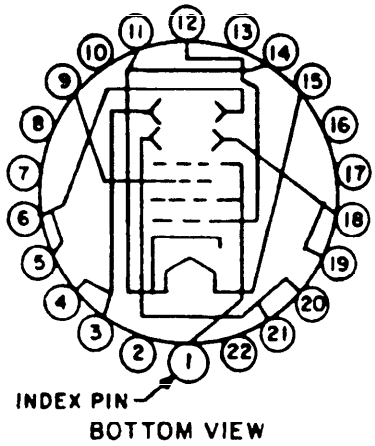
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22A



BOTTOM VIEW

Pin no.		Element
1	-	Accelerator
3 & 4	-	Deflecting electrode D_1
5 & 6	-	Deflecting electrode D_2
9	-	Focusing electrode
11 & 14	-	Heater & cathode
12 & 13	-	Grid no. 1
15	-	Heater
18 & 19	-	Deflecting electrode D_3
20 & 21	-	Deflecting electrode D_4

NOTES:
a. Minimum useful screen diameter.
b. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 1GALP19.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note b)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	10.375 (263.53) Dia.	10.625 (269.88) Dia.
B	9.000 (228.60) Dia.	
F	11.375 (288.93)	11.875 (301.63)
G	15.500 (393.70)	16.000 (406.40)
H	19.750 (501.65)	20.250 (514.35)
J	3.469 (88.11)	3.656 (92.86)
K	1.594 (40.49)	1.969 (50.01)
L	4.312 (109.52) Dia.	4.500 (114.30) Dia.
	Reference dimensions (see note c)	
C	42.000 (1066.30) Rad.	
D	11.750 (298.45) Rad.	
E	1.000 (25.40) Rad.	

FIGURE 1. Outline drawing of electron tube type 10ALP19 - Continued.

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